Intermolecular Forces

Dispersion, dipole–dipole and hydrogen bonding

Name _____ Class _____ Date _____

DATA

Hydrogen bonding: H on N, O or F + lone pair on N, O or F

ΔH(vap) at the boiling point: water 40.7 kJ/mol · ethanol 38.6 kJ/mol · methane 8.19 kJ/mol

Molar masses (g/mol): H_2O 18.02 · C_2H_5OH 46.07 · CH_4 16.04

Show units on every quantity in every step.

A. Understanding

1. (a) What is an intermolecular force? (b) Explain why boiling water does not break O–H bonds.

B. Identifying forces

2. List all the intermolecular forces in: (a) Ar (b) CH_3Cl (c) CH_3OH (d) CCl_4 (e) HF (f) H_2S

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C. Boiling points

3. The boiling points of F_2, Cl_2, Br_2 and I_2 are −188, −34, 59 and 184 °C. Explain the trend, and give the state of each at 25 °C.

4. Pentane boils at 36 °C and 2,2-dimethylpropane at 10 °C. Both are C_5H_{12}. Explain the difference.

5. For each pair, predict which has the higher boiling point, and explain: (a) ethanol, CH_3CH_2OH, or methoxymethane, CH_3OCH_3 (b) HCl or HBr (c) NH_3 or PH_3

6. Explain why ice floats on liquid water.

D. Solubility

7. Predict whether each dissolves better in water or in hexane, and explain: (a) I_2 (b) KCl (c) ethanol (d) octane

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E. Calculations

8. Calculate the energy needed to vaporize (a) 100. g of ethanol (b) 100. g of water, each at its boiling point. (c) Why is the answer to (b) much larger?

9. Calculate the enthalpy of vaporization per gram of methane and of water. How many times more energy per gram does water need? Explain in terms of intermolecular forces.

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